Unit 11, Lesson 2: Constant of Proportionality Part 1

Benchmarks

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

Learning Targets

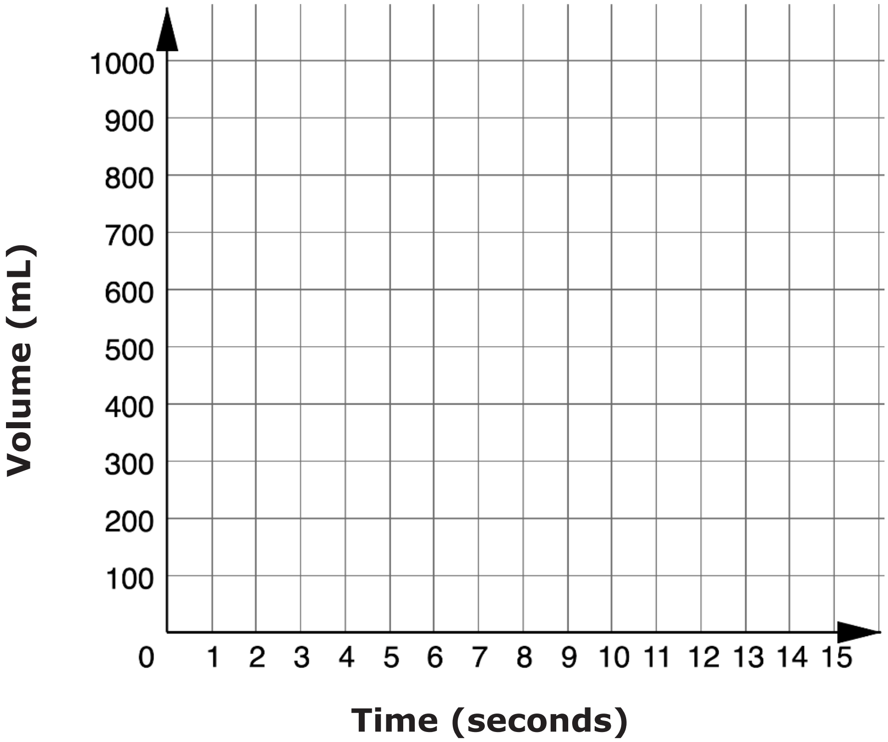
- ☐ I can graph a proportional relationship from a table, an equation, or a written description.
- ☐ I can determine the constant of proportionality from a graph.
- ☐ I can determine the constant of proportionality from a table.

11.2.1 Warm-Up: Graphing Story

A student is pouring water from a container into an empty beaker. The beaker measures volume in milliliters (mL). The screenshots show the volume of water in the beaker at different points in time. A line is sketched at the approximate water level in each screenshot.

Volume at 4 Seconds	Volume at 10 Seconds
	

1. Sketch the graph of the relationship.



11.2.2 Exploration: Unit Rates and the Graphs of Proportional Relationships

Work with your partner to complete the following.

1. Malcolm went to the gas station to fill up his tank with gasoline. The current price for regular unleaded gasoline at the station is $\$2.74$ per gallon.

   a. Determine an approximately equivalent ratio to represent the same relationship per dollar. Round to the nearest thousandth.

$$\frac{\$2.74}{1\text{ gallon}} \approx \frac{\$1}{\boxed{}\text{ gallons}}$$

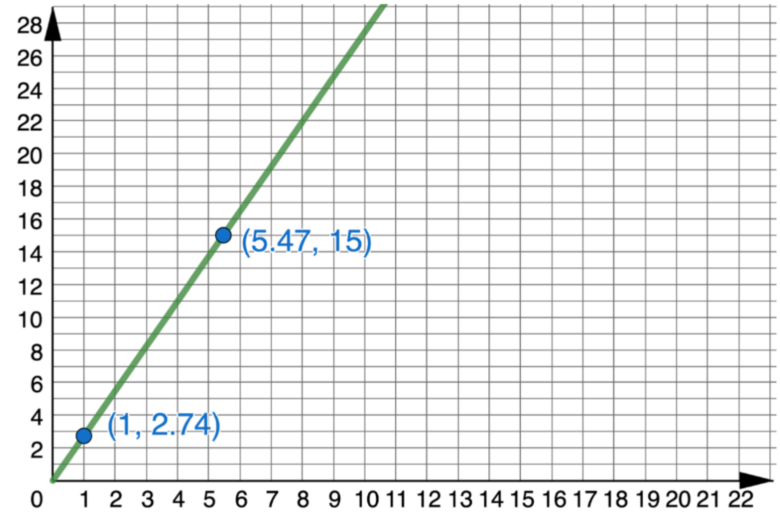
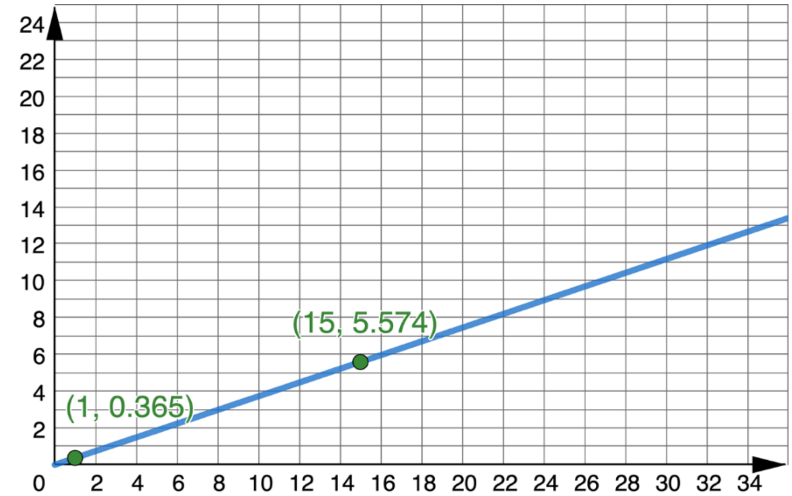
   b. Both $\frac{\$2.74}{1\text{ gallon}}$ and $\frac{\$1}{\boxed{}\text{ gallons}}$ are unit rates. Complete each statement to interpret both unit rates.

   - Each gallon of gasoline will cost Malcolm \$________.
   - For every $\$1$ Malcolm spends, he can add ________ gallons to his tank.

   c. Use either unit rate $\frac{\$2.74}{1\text{ gallon}}$ or $\frac{\$1}{\boxed{}\text{ gallons}}$ to complete the table of values to represent this situation. Round to the nearest hundredth when necessary.

Gallons of Gasoline	12		2.5	
Cost in Dollars		22.74		15

Two different graphs can be used to represent the proportional relationship between gasoline, in gallons, and cost, in dollars. Both graphs are shown.

Graph A	
Graph B	

d. Label the axes of both graphs based on your work in Parts A-C.

e. Discuss with your partner which point on each graph represents a unit rate for this relationship. Then, place an X on each graph at this point.

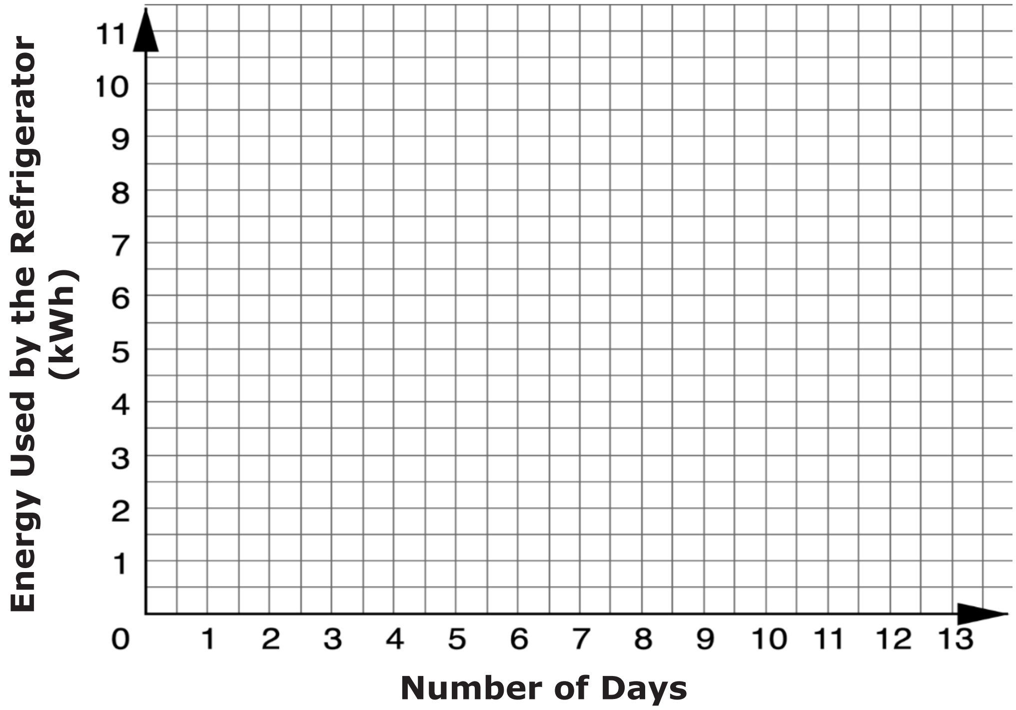
11.2.3 Guided Instruction: Constant of Proportionality

1. Ana's refrigerator uses an average of 1.6 kilowatt-hours (kWh) of energy each day.

   a. Complete the table of values to represent this relationship.

Number of Days	0	1	2	3
Energy Used (kWh)				

   b. Plot the ordered pairs from the table on the coordinate grid shown.



Complete the statements to describe the relationship between each point plotted on the graph in Part B.

The y-value ○ increases ○ decreases by ○ 1 unit ○ 1.6 units every time the x-value ○ increases ○ decreases by ○ 1 unit ○ 1.6 units

   c. Use this information to determine two additional points on the graph of this relationship.

The graph of the relationship can be completed by using a straightedge to draw a ray with an endpoint at $(0,0)$, going through each point that represents the relationship between the two quantities, with an arrow extending beyond the grid shown.

d. Complete the graph to represent the relationship by drawing the ray.

e. Discuss with your partner why the graph of the relationship representing this context has an endpoint at $(0,0)$ and a ray that extends beyond the grid shown.

The graph of a proportional relationship represents an infinite number of points. The point $(3.75,6)$ is included on the graph.

ABC

The ***constant of proportionality*** is the constant value of the ratio of two proportional quantities.

f. Explain what this point means in terms of the context.

g. Place an X at this point on the graph.

From a graph or a table of values representing a proportional relationship, the constant of proportionality can be determined by expressing any ordered pair as a ratio.

The **constant of proportionality** is the unit rate. It can be determined using the ratio $\frac{a}{b}$ or $\frac{b}{a}$. When two quantities are in a proportional relationship, there are two constants of proportionality, and they are always reciprocals of each other.

2. Two points on the graph and in the table of values representing the proportional relationship between the number of days and the amount of energy Ana's refrigerator uses are $(3, 4.8)$ and $(2, 3.2)$.

a. Complete each ratio of the energy used, y, to the number of days, x, for each ordered pair.

$$\frac{y}{x} = \_\_\_\_\_ = \_\_\_\_\_$$

b. Determine the value of each ratio using a calculator.

c. Complete the statement.

The constant of proportionality is the value of each ratio in this relationship, which is _____.

d. Find the reciprocal of the constant of proportionality from Part C.

e. Complete the statements.

The other constant of proportionality for the same relationship is _______. In this context, this means as the

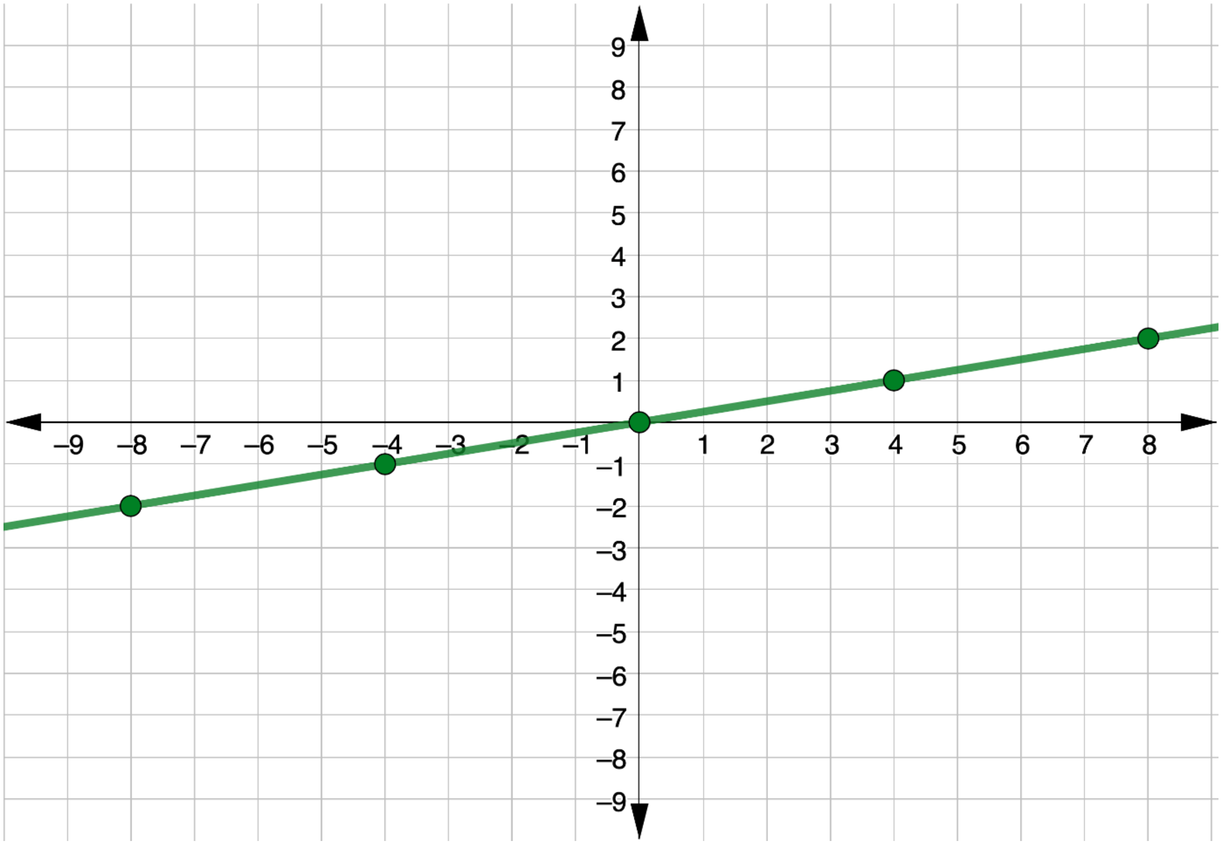
- ○ number of days the refrigerator runs
- ○ amount of energy the refrigerator uses

increases by _______ units for every 1 unit increase in the

- ○ number of days.
- ○ amount of energy used.

11.2.4 Guided Practice: Constant of Proportionality

1. The graph of a proportional relationship is shown.



a. Determine the constant of proportionality, $\frac{y}{x}$.

b. Complete the statement to describe the constant of proportionality.

The y-value ○ increases ○ decreases by _______ every time the x-value increases by 1.

2. The table of values shows how many tablespoons (tbsp) of chocolate milk Dax adds to different quantities of milk to make his preferred flavor of chocolate milk.

Cups of Milk, c	**Tablespoons of Chocolate Syrup, T**
$\frac{1}{6}$	$\frac{1}{2}$
$\frac{1}{3}$	1
1	3
$1\frac{1}{2}$	4.5
2	6

a. Determine the constant of proportionality, $\frac{T}{c}$.

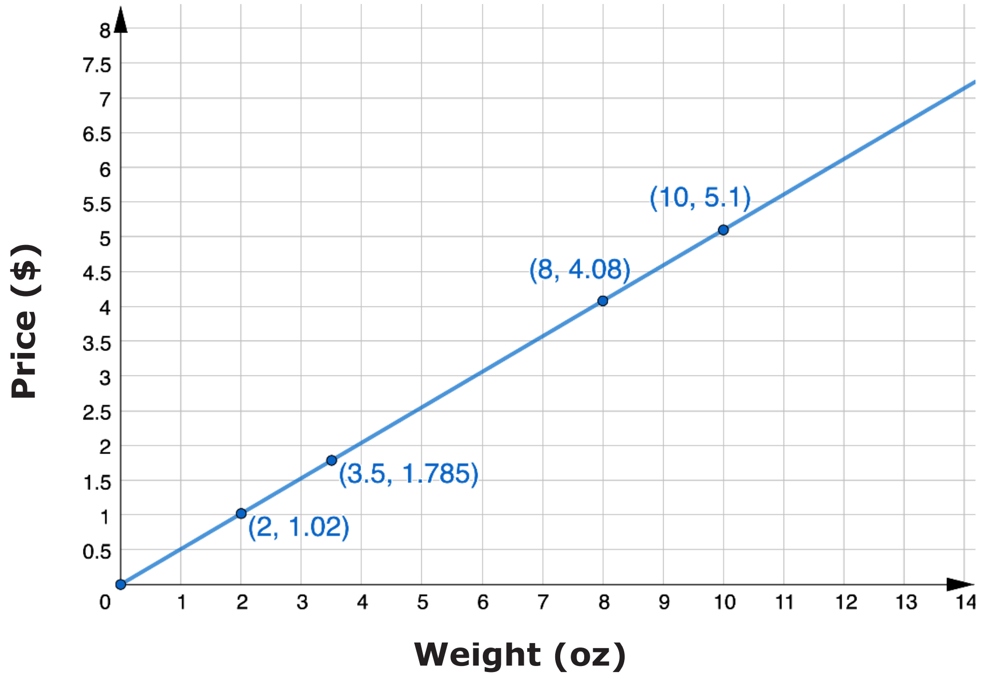
b. Complete the statement to interpret this constant of proportionality.

Dax [○ increases ○ decreases] the amount of [○ syrup ○ milk] used by _______ [○ cups ○ tablespoon] every time the amount of [○ syrup ○ milk] used increases by [○ 1 cup. ○ 1 tbsp.]

c. Determine the constant of proportionality, $\frac{c}{T}$.

11.2.5 Your Turn!

1. At a local frozen yogurt restaurant in Lake Mary, FL, the price of frozen yogurt is determined using its weight, in ounces (oz). The graph shows this relationship.



a. Determine the constant of proportionality, $\frac{y}{x}$.

b. Complete the statement to describe the constant of proportionality.

The ○ price ○ weight ○ increases ○ decreases by __________ ○ dollars ○ ounces every time the ○ price ○ weight increases by ○ 1 dollar. ○ 1 ounce.

11.2.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept after today's lesson.

I need help. I got it. I could help someone.

Understand what the constant of proportionality is in a proportional relationship.	
Determine the constant of proportionality from a graph.	
Determine the constant of proportionality from a table.	